PROJECT TECHNICAL DOCUMENTATION

BioSpecInfo

Web platform for computational chemistry, physics and mathematics

Release Conformance Statement

DOCUMENT CONTROL

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Title	Release Conformance Statement
Software version	**bsi-v248**
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Document generated from the versioned source in the project repository. The version and revision shown here and in the footer of every page identify exactly the state of the software described.

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Release Conformance Statement — BioSpecInfo

Field	Value
Software	BioSpecInfo
Version	`bsi-v248`
Author and release owner	Samuele Pio Provenzano
Repository	`github.com/samupropio1-ship-it/BioSpecInfo-v11`
Distribution	GitHub Pages — `samupropio1-ship-it.github.io/BioSpecInfo-v11/`
Nature of the software	Client-side Progressive Web App, no backend

1. Subject of this statement

The undersigned declares that version `bsi-v248` of BioSpecInfo has been put through the verification procedure described in `02-Verification-Validation-Report.md` and that the outcomes are those reported, without selection, in `../evidence/RAPPORTO-VERIFICA.md` — a document generated automatically by `tools/genera-evidenza.js`, which transcribes the benches' output with no manual intervention.

This statement does **not** attest the absence of defects. It attests that:

1. an automated verification procedure exists and can be reproduced by third

parties;

1. the outcome reported is the one actually produced, failures included;
2. the known limitations are declared explicitly and not presented as features.

2. Intended use and limitations

2.1 Declared purpose

BioSpecInfo is a **teaching and study-support** tool for chemistry, biochemistry, pharmacology and astrochemistry. It is designed for university students and for exam preparation.

2.2 What the software is NOT intended for

These exclusions are an integral part of this statement.

Domain	Exclusion
Clinical use	It is not a medical device within the meaning of Regulation (EU) 2017/745. It does not support diagnosis, prescription, dosing or therapeutic decisions. The pharmacological data are for teaching purposes only, and the application says so at

Domain	Exclusion
	the head of the section.
Certified analytical use	The spectra are **predictions** from functional-group tables, not measurements nor quantum-mechanical simulations. They do not replace an experimental database (SDBS, NIST) nor an instrumental determination, and cannot be used for the formal identification of substances.
Regulated GxP environment	It is not validated for use in a GLP/GMP environment. The alignment with GAMP 5 described in `docs/03` is methodological, not a formal qualification.
Access control	The File Manager protects its contents with a client-side deterrent. On a static site there is no effective access control: whoever reads the source bypasses it. It must not be used for confidential material.

3. Summary of verification outcomes

The full outcomes, with each bench's textual output, the execution environment, the commit and the SHA-256 digests, are in `../evidence/RAPPORTO-VERIFICA.md`.

Family	What it demonstrates
Scientific data	The chemical data are verified against an independent source
AI agent	The assistant stays usable when the external provider fails
Stability	The app withstands long sessions, exhausted storage, degraded networks
Interface	The panels respond as documented
Documentation/code consistency	What the documentation promises exists in the code

4. Known deviations at the date of release

Listed in full. None has been removed from verification in order to make it pass.

#	Deviation	Impact	Treatment
D-01	**2 entries** (it was 6) that are not single molecules: Ivermectin, a mixture of B1a/B1b homologues, and Coartem, a combination of two active ingredients	For these two no 2D/3D representation and no spectral prediction is available	Entries left **without SMILES**: the pharmacological data remain, the structure is not shown. For Digoxin, Vincristine and Tacrolimus (topical and systemic) the structure was taken from **ChEMBL** and passes the comparison with the declared molecular weight: those four left the registry. The two that remain have no structure to show, not a missing one.

#	Deviation	Impact	Treatment
D-02	19 entries with no SMILES by their very nature (monoclonal antibodies, peptides)	None: for those molecules SMILES notation is not the appropriate representation	Not a substantive deviation; listed for completeness
D-03	~~SEC-02 verified only by proxy~~ — **remedied**: see S-10	—	`audit_rete` observes the traffic during real use, with a canary value seeded into the user's own data
D-04	~~The File Manager password in the git history~~ — **remedied**: see S-12	—	The password was **changed** at version `bsi-v188`. The old one is still readable in the history and no longer opens anything. At the author's request it was changed **again** at `bsi-v245` (only its SHA-256 fingerprint is in the code)
D-06	Code coverage **measured but partial**: 49.89 % of statements, not branches	It is the coverage of the widest path a bench walks (194 sections plus the other pages), not of the whole battery; and an `if` entered from one side only counts as covered	`audit_copertura` measures it with Chromium's profiler, with no build and without rewriting the source. The value is **recorded**: if it falls, the battery fails
D-07	Verification on Chromium only	Firefox and WebKit are tested by hand	Declared in `docs/08` §8. In the verification environment the reason is checkable: the CDN from which `playwright-core` downloads the other engines answers **403** to the network policy, so Firefox and WebKit cannot be installed there. **Re-verified on 2026-10-08**: `npx playwright install firefox` fails with «Download failure», and `cdn.playwright.dev` still answers 403 to the tunnel. It is not a choice, it is a constraint of the environment, and it stands
D-08	~~Partial English translation~~ — **remedied**: see S-13	—	All **16 documents** are in English. `verifica-documenti` and `verifica-affermazioni` read `docs/en/` just like the Italian set: a disagreement between the two languages fails the battery

#	Deviation	Impact	Treatment
D-09	Full WCAG 2.1 AA conformance not verifiable entirely by automation	What stays outside is text inside SVGs (8,888 elements), text over a real background **image** (108) and everything requiring human judgement. Text over a **gradient** has entered the measurement: 591 elements, judged against the worst stop of the gradient	The skipped elements are **counted** and reported on every run
D-10	The `bsi-v181` tag points at the wrong commit	The release workflow, started without filling in the version field, used its own default — `bsi-v181` — and, being rerunnable by design, **moved that tag** from commit `cb0cf03` to the `bsi-v188` one, also attaching the v187 packages to it. The foreign assets were removed. **The tag cannot be moved back by an app**: GitHub refuses to let an app write a ref pointing at a commit whose `.github/workflows/*` differs from the current one, and at `cb0cf03` that file differs by 319 lines. Three routes were tried, all blocked by the same rule: `git push --force` («refusing to allow a GitHub App to create or update workflow»), `PATCH /git/refs` with `force` (403), and delete-and-recreate — where the **403 already lands on creating a throwaway tag** at that commit, so the real tag was left untouched. `GITHUB_TOKEN` cannot be granted the `workflows` permission. **Re-verified on 2026-10-08** with the session's own GitHub client, which uses a different credential: `PATCH /git/refs/tags/bsi-v181` answers **403 «Write access to this GitHub API path is not permitted through this proxy»**. It is a second block, independent of the first: even without the workflow-file rule, writing to git refs does not get through. The tag points at `64e8c316`; it should point at `cb0cf03e`	The three packages attached to the release are the correct ones; the «Source code (zip)» archive must be ignored, and the `bsi-v181` code is downloadable from commit `cb0cf03`. Declared in the CHANGELOG's `bsi-v181` entry and in the release body, with the command that restores the tag from personal credentials. The default was removed and the workflow now **refuses** a version that does not match `sw.js` and the package filenames — verified both ways
D-11	Part of Spectra Live, the per-subject ✨ menu and the quick tools have no bench	Voice, continuous listening, local commands other than the creator's, the tools that act on the page and the physics and mathematics ✨ menu are tested by hand: a failure in them would not fail the battery	Declared in `docs/08` §6 as **UI-21** and **AI-13**. Since v238 the console layout, the navigator, the level, the face and the creator commands are covered by `test_spectra238` (UI-22, AI-14, SEC-08); since v239 the guided lesson and the «fammi lezione» command by `test_lezione` (UI-24) and the single button by `test_spectra238` (UI-25), while the lesson's voice is still tested by hand. Speech recognition and synthesis do not exist in the battery's headless Chromium;

#	Deviation	Impact	Treatment
			the quick tools' computations rest on `BSIMatCAS`, verified by `test_matematica`
D-12	Two Spectra local keys stay outside selective erasure	`bsi-spectra-ritratto-v4` (the icon derived from the face) and `bsi-spectra-sentinella` (continuous-listening mark) are not in the «Clear data» groups. The mind, the creator recognition, the subject and the study kits are (Persistent memory and Progress groups)	Neither holds user data: the first is regenerated, the second is removed when listening stops. Declared in `docs/11` §6
D-13	Spectra's creator recognition is not an access control	The SHA-256 fingerprint of the secret phrase is in the code, and the recognition mark (`bsi-spectra-creatore-v1`) is in `localStorage`, where whoever holds the device can write it by hand. An unsalted fingerprint of a weak phrase can also be searched by trial outside the app	Like the File Manager password (SEC-03), it is a deterrent: it protects no data, it tells Spectra whom it is talking to. The phrase is neither in the code nor stored (SEC-08, `test_spectra238`)
D-14	Spectra with the body: the body is included in the app, but **packed**, not as a `.glb` (since v241)	The body chosen by the author (the «Bounty Hunter», free on CGTrader) is under CGTrader's Royalty Free licence: it may be used inside a product, not offered as a standalone file. Up to v240 it was not in the app and each user had to load it; since v241 it is always there, for everyone, as the packed block `models/spectra-corpo.bsi` (4.40 MB, «BSIC» format: the `.glb` bytes mixed with an xorshift32 stream from a fixed seed) that only the app's code reopens. **It is not absolute protection**: someone reading the code can rebuild the `.glb`, as with any model shown on a screen	It is how games and apps embed purchased models: the published file is not a downloadable, reusable `.glb`; the original `.glb` stays out of git (`.gitignore`). The block is regenerated with `tools/impacchetta-corpo.js`, which checks that it reopens identical. Rig, figure and reactions are tested on the **real embedded body** (`test_spectra_studio`, `test_spectra_figura`: AI-17, UI-26, UI-27); the visual rendering has to be judged by looking

Contrast, from 1,069 defects to zero — how it surfaced and how it was closed

Up to version `bsi-v168` this document declared WCAG conformance "verified by sampling" and the bench reported **0 defects**. Those were two statements at once true and meaningless: the bench inspected only the **visible** elements, and `index.html` shows one section at a time. Out of 19,751 text elements it was looking at **41**.

Walking all 87 sections then present, **1,069** contrast defects surfaced. They had not appeared: they had always been there.

Today they are **zero**, measured with the same bench over every section — 88 of them now. Not by wholesale substitution — that road, attempted once, introduced 997 new defects — but by going back to the **common cause** every time and re-measuring after each change. The **40 unlabelled fields** have likewise fallen to **zero**.

The gravest causes were not wrong colours but **theme tokens whose value equalled the background**:

Structural defect	Measured effect
`--g800` is `#0d1522`, **identical to** `--bg` — and the rule `a { color: var(--g800) }` applied it to every link	Every link without a colour of its own was at contrast **1:1**: invisible
`--g700` is `#16263d`, nearly identical to `--white` (`#16273e`)	Text at **1.01:1**
Buttons that changed their **background** without changing their **text colour**	The selected state (or the deselected one, depending on the panel) became unreadable: **1.39:1**
`lightenColor(hex,pct){ return hex; }`	A function called "lighten" that returned the colour **unchanged**: an unfinished stub, in all likelihood written for this very problem
The quiz answer panel: light background, light text	The student **could not read the correct answer**

The structural remedy was an explicit token — `--testo-forte` — for prominent text on a dark surface, applied only to the uses as a *text colour*: `--g800` and `--g700` stay where they serve as background or border.

Cause	Defects	Remedy
A dark-theme safety net with higher specificity than the components: text meant for the **white** cards was forced to `#d4dce6`, contrast 1.38:1	159	Rewritten with `:where()` (zero specificity): it protects text that has no colour of its own and stops fighting text that has one
A per-category accent used as **text** on a dark header, with `#0d1522` (near-black) as fallback: 1.13:1	~400	`bsiAccentoLeggibile()` preserves the hue and raises the lightness just enough. Applied at the 14 points where the accent is chosen — not in the data, so it holds for future entries too
Two hues used only on dark backgrounds, just under threshold (3.75:1 and 3.04:1)	139	Replaced after **verifying** that they never appeared on a light background
Table zebra striping white/near-black with always-light text: the white rows were unreadable	57	Two dark tones, consistent with the theme
Periodic-table cells just under threshold (4.46:1 and 3.88:1)	43	Text at `#eceff4`, verified across all nine categories
`"#var(--text)"` — one `#` too many makes the colour invalid: in SVG it falls back to black, on canvas the assignment is **ignored** and the previous colour persists	8 places	Replaced with the real value

The direction of the correction depends on the surface, not on the colour. The first attempt always lightened: it removed thirty defects on the dark header and added eighteen on the light formula box that uses the same accent. The function now darkens on light backgrounds and lightens on dark ones.

The final tail: 80 → 0. The last eighty were scattered over 33 distinct causes and looked like finishing work. They were not: three more structural causes lay underneath.

Final cause	Remedy
The "automatic contrast corrector" was producing defects instead of removing them. It used *perceived* brightness instead of WCAG luminance, so its thresholds corresponded to no real ratio; it **ignored alpha**, so `rgba(255,180,84,.08)` over a dark surface looked to it like a "light background" and it darkened the text to `#16273e` — **creating** a defect at 1.21:1 on top of a source that was already correct; and it looked only at elements carrying a background of **their own**, that is, never the commonest case	Rewritten: it walks the ancestors **compositing alpha** to obtain the effective background, measures the true WCAG ratio, and delegates the correction to `bsiAccentoLeggibile()`
`--g900` is `#0d1522` — **the same as `--bg`**, exactly like `--g800`: the legacy of a greyscale born for the *light* theme, where `--g900` was "the darkest text". With the theme inverted, that role no longer exists	The uses as a **text colour** (30 rules today) move to `--testo-forte`; the 16 uses as a **background** stay where they are
A colour chosen from data used as a **background** with the text hard-coded to `#fff`: it works while the hue is dark and stops as soon as someone adds `#e65100`	`bsiEtichettaLeggibile()` — the dual of the previous function: it picks the text by looking at the background and, if neither extreme suffices, **darkens the background while preserving the hue**, because a label must stay recognisable by colour

The steps as measured with the official bench, not estimated: **80 → 27 → 26 → 10 → 4 → 0**. Each step is a full re-measurement over the 87 sections after a group of changes; none of the figures above is attributed to a single cause, because the causes were not measured one by one.

The number remains **recorded as a baseline** in `evidence/accessibilita-riferimento.json`. Zero is not an achievement to file away: it is the value the bench defends. If a future change reintroduces even one defect, the battery **fails**.

What stays outside automation is what automation cannot judge: text inside SVGs and text over gradient backgrounds (D-09). Those elements are **counted** and reported on every run, not passed over in silence.

4-bis. Deviations remedied in this version

They are listed because they concerned **the verification apparatus itself**, and a conformance statement that says nothing about the defects of its own evidence is not worth the paper it is written on.

#	Deviation remedied	Why it mattered
S-01	32 of the 39 benches were not in the repository: they lived in the working area of the session in which they had been written	It made **false** the sentence this whole document rests on — "anyone can re-run them". Whoever cloned the repository found 7 out of 39. They now live in `tools/banchi/`
S-02	A missing bench was counted but did not make the battery fail	A battery missing *all* its benches would have declared "CONFORMANT" with zero failures. A missing bench now makes the outcome **NON-CONFORMANT**
S-03	`package.json` and `package-lock.json` were excluded from the repository	`npm install`, the documented command for reproducing the evidence, installed nothing: the battery failed at the first bench with *Cannot find module*
S-04	The accessibility check inspected only the visible section	On `index.html` that meant one section out of 87. The "0 defects" held for the Dashboard, not for the application
S-05	`SCI-10` was defined twice in the matrix, and the coverage totals were wrong (36 declared out of 37 defined)	The first number an assessor reads was wrong. A bench now verifies it
S-06	50 broken links out of 127 in the delivery packages	Documentation handed to third parties with references that led nowhere
S-07	15 attachable PDFs stuck at documents 00-05 and at an earlier version	An obsolete attachment states false things with an air of authority
S-08	The numeric-claims bench looked at **the Italian documents only**	The English set drifted undisturbed: `docs/en/05` declared "84 sections" while the app has 87, and stayed stuck at version `bsi-v146` while the code was at 171. The English set is now compared against the measurement just like the Italian one, and a disagreement between the two languages fails the bench
S-13	The English translation stopped at 10 documents out of 16	A non-Italian-speaking assessor read less than two thirds of the documentation, and the missing part was the operational one: data model, interfaces, deploy, functional specifications, manual. All translated: the English set is now as complete as the Italian one, and the two consistency benches check it the same way
S-12	The File Manager password had been left in clear text in the git history	Taking it out of the files does not take it out of the history: `git log -p` hands it to anyone, and as long as that password is in use the deterrent deters nobody. The only remedy is to **change it**, and that has been done: the source carries only the new SHA-256 digest, the old password remains in the history and no longer opens anything. Verified in the browser: the new one opens, the old one does not

#	Deviation remedied	Why it mattered
S-10	SEC-02 — "no personal data leaves the device" — was verified by checking the two *mechanisms* of exit, not the exit	A check on mechanisms says "I cannot see how it would leave", which is not "it did not leave". `audit_rete` seeds an unrepeatable value into 71 stores of the user's data, then **uses** the application and inspects the URL, headers and body of every request. Proved in reverse with `BSI_PROVA_FUGA=1`, which causes a leak on purpose: the bench must fail, and it does
S-11	The deviations registry had no converse guard: an entry declared that **no longer deviates** simply stayed	An exemption nobody revokes is a permission left switched on, and tomorrow it would silently cover a wrong structure put in its place. The bench now fails for that too — verified by putting a remedied entry back
S-09	The "automatic contrast corrector" **created** the defects it was meant to remove	It ignored alpha: a `rgba(255,180,84,.08)` background over a dark surface looked light to it, and it darkened with `!important` a text that was already correct in the source. A remedy tool that worsens the thing it remedies is the hardest kind of defect to see, because it presents itself as the solution

5. Acceptance criteria applied

A version is published only if **all** the following criteria are satisfied.

- ☑ No non-network JavaScript error on any page
- ☑ `tools/verifica-farmaci.js` — no defect; every deviation recorded
- ☑ `verifica_guida` — no promise in the documentation without a counterpart in the code
- ☑ No conflict marker in the tracked files
- ☑ Service Worker cache version bumped and aligned with `BSI_APP_VERSION`
- ☑ Known deviations listed in §4 of this document
- ☑ Evidence report regenerated against the published commit
- ☑ **No missing bench** — a bench that is not there is not a bench that passed
- ☑ `tools/genera-pacchetti.js` closed with **0 broken links**
- ☑ `tools/genera-pdf.js` re-run: the attachments describe this version

6. Change management

Aspect	Procedure
Traceability	Every change goes through a pull request describing the defect, its cause and its verification
Versioning	The application version coincides with the Service Worker cache version (`bsi-vNNN`), incremented at every publication
Distribution	Automatic from `main` via GitHub Pages; the Service Worker signals the new version to open sessions without interrupting them

Aspect	Procedure
Regression	The full battery is re-run before every merge into `main`
Reversibility	Every version corresponds to a commit; rollback is a `git revert`

7. Intellectual property and licences

All third-party components are distributed under **permissive** licences (MIT, BSD-3-Clause, Apache-2.0, CC0, Public Domain), mutually compatible and free of strong copyleft constraints. The «Noir» 3D model used as Spectra's face (`models/spectra-noir.glb`) is by Skyfox13 (<https://sketchfab.com/podderkuntal25>), under the CC BY 4.0 licence, and has been modified; the attribution appears in the Spectra Live console, in the README, in `THIRD_PARTY_NOTICES.md` and in `07-SBOM.md`. The full bill of materials, with digests and residual obligations, is in `07-SBOM.md` and in CycloneDX format at `../evidence/sbom.cdx.json`.

The code written for this project is the author's own work.

8. Data handling

The application is **local-first**: there is no backend, no personal data is collected and no persistent identifier is present. Data generated by the user (notes, chats, API keys, uploaded files) stay in the device's `localStorage` and IndexedDB and can be deleted selectively from the interface itself.

Spectra's mind (`bsi-spectra-mente.js`) keeps conversation episodes, study time and an internal state in `localStorage`: it is a simulation of behaviour, not consciousness. When a question goes to the artificial-intelligence provider the user chose, a summary of this memory and of the open tab goes with it; if the question is about what is on screen, an image of its canvases too.

The creator recognition (`bsi-spectra-creatore.js`) keeps only a mark (`bsi-spectra-creatore-v1`: recognised yes, date of first and last access). The secret phrase is compared as a SHA-256 fingerprint and enters neither the dialogue, nor memory, nor voice, nor the requests to the provider; the model receives only whether the user has been recognised.

The guided lesson (`bsi-lezione.js`) is built from the open page: it does not call the model, has no keys of its own in `localStorage` and goes online only for the voice, as below; at the end it notes an emotion («fiera», proud) in Spectra's mind, which ends up in `bsi-spectra-mente-v1` like the others.

Spectra's quality voice (`bsi-spectra-voce.js`, since v239): if the user has saved an OpenAI key, the text Spectra speaks in the console, in the guided lesson and in Spectra Studio is sent to OpenAI (`gpt-4o-mini-tts`) to receive the audio, even when the chat uses another provider and without going through the proxy. Without an OpenAI key the browser's speech synthesis is used and the text does not leave. Spectra Studio (`spectra-studio.html`) keeps in

`localStorage` the predictions and the exercises opened for each lesson (`bsi-spectra-studio-v1`) and the voice choice (`bsi-spectra-studio-voce`), and also sends each prediction to the mind. Both keys are covered by «Clear data → Persistent memory». With the voice off in the studio (🔇) no text goes to OpenAI. Since v241 Spectra's body is embedded in the app (`models/spectra-corpo.bsi`): the user no longer loads any file and nothing is written to IndexedDB; an old body stored by a previous version (`bsi-spectra-modelli`) is deleted on start, and «Clear data → Progress in the rest of the app» deletes that database.

Network calls are optional and go to documented public services (§4 of `07-SBOM.md`). If the user configures the proxy, the API keys reside on a Worker they own and not on the device.

Details in `03-Security-Privacy-Compliance.md`.

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9. Signature

This document is drawn up by the author of the software, who takes responsibility for it. It does not constitute certification by a third-party body, and is not presented as such.

Anyone can verify what is declared here by re-running the procedure in §5 of `../evidence/RAPPORTO-VERIFICA.md` against the commit indicated, and comparing the SHA-256 digests of the files.

Samuele Pio Provenzano *Version* `bsi-v248`.